

Lab 11

Case Study

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To develop a GIS-based tool that can monitor drought conditions using vegetation and temperature indices.

Tool Properties: DMT

General

Parameters

Execution

Validation

Environments

Label: DroughtMetrics

Toolbox: E:\M.Sc_II Sushant\GIS Programming\Lab_9\Lab_9\Task.atbx

Description: A GIS-based tool for automated drought assessment that calculates vegetation and temperature indices using satellite imagery to monitor drought conditions effectively.

Summary: Tool designed for efficient drought monitoring and assessment using ArcGIS Pro and ArcPy. By processing satellite imagery bands—specifically Red, NIR, Green, and optionally Land Surface Temperature (LST)—the tool calculates essential drought and vegetation indices such as the Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI), Normalized Difference Moisture Index (NDMI), Vegetation Condition Index (VCI), and Vegetation Health Index (VHI). If LST data is provided, it also calculates the Temperature Condition Index (TCI) to further enhance drought analysis.

[Learn more about script tools](#)

OK Cancel

Tool Properties: DMT

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	Label	Name	Data Type	Type	Direction	Category
0	Red Band	Red_Band	Raster Layer	Required	Input	
1	Near Infra...	Near_Infrared_(NIR)_Band	Raster Layer	Required	Input	
2	Green Band	Green_Band	Raster Layer	Required	Input	
3	Land Surf...	Land_Surface_Temperature_(LST)_Band	Raster Layer	Optional	Input	
4	Clip Shap...	Clip_Shapefile	Feature La...	Optional	Input	
5	Output Fo...	Output_Folder	Workspace	Required	Input	
*			String	Required	Input	

[Learn more about script tools](#)

OK Cancel

Tool Properties: DMT

General

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Script File   ... embedded ...

```
import arcpy
from arcpy.sa import *

# Set input parameters
red_band = arcpy.GetParameterAsText(0)      # Red band (required)
nir_band = arcpy.GetParameterAsText(1)      # NIR band (required)
green_band = arcpy.GetParameterAsText(2)     # Green band (required)
lst_band = arcpy.GetParameterAsText(3)      # LST band (optional)
clip_shapefile = arcpy.GetParameterAsText(4) # Clip area shapefile (optional)
output_folder = arcpy.GetParameterAsText(5)  # Output folder

# Set output paths
ndvi_output = output_folder + "/NDVI.tif"
ndwi_output = output_folder + "/NDWI.tif"
ndmi_output = output_folder + "/NDMI.tif"
vci_output = output_folder + "/VCI.tif"
vhi_output = output_folder + "/VHI.tif"
tci_output = output_folder + "/TCI.tif"

# Check out the Spatial Analyst extension
arcpy.CheckOutExtension("Spatial")

# Clip raster inputs if a shapefile is provided
def clip_raster(raster):
    if clip_shapefile:
        return ExtractByMask(Raster(raster), clip_shapefile)
    return Raster(raster)

# Clip the bands
red_band = clip_raster(red_band)
nir_band = clip_raster(nir_band)
green_band = clip_raster(green_band)
lst_band = clip_raster(lst_band) if lst_band else None

# Calculate NDVI
ndvi = (nir_band - red_band) / (nir_band + red_band)
ndvi.save(ndvi_output)

# Calculate NDWI
ndwi = (green_band - nir_band) / (green_band + nir_band)
ndwi.save(ndwi_output)

# Calculate NDMI (requires SRTM for red band - replace with your data)
```

Geoprocessing

←

DroughtMetrics

+

Parameters

Environments

?

Red Band

LC08_L2SP_146048_20231117_20231122_02_T1_

Near Infrared (NIR) Band

LC08_L2SP_146048_20231117_20231122_02_T1_

Green Band

LC08_L2SP_146048_20231117_20231122_02_T1_

Land Surface Temperature (LST) Band

LC08_L2SP_146048_20231117_20231122_02_T1_

Clip Shapefile

jat.shp

Output Folder

Tool_Outputs

Run

✓

DroughtMetrics completed.

View Details

Open History

Suggestions

Catalog

Geoprocessing

Drawing Order

Map

✓

VHI

Value

88.1476

13.3544

✓

TCI

Value

100

0

✓

VCI

Value

100

0

✓

NDMI

Value

0.443834

-0.0525388

✓

NDWI

Value

0.022063

-0.405573

✓

NDVI

Value

0.443834

-0.0525388

